SET 3 | SYSTEM ENGINEER

Infosys
Numerical Puzzle

Set 3: 30 New Practice Questions

All-new puzzles with answers and step-by-step solutions.

30	4	100%
Questions	Options each	With solutions

WHAT'S INSIDE

Number Series
8 questions
Interleaved, power and difference patterns

Grids & Magic Squares
5 questions
Row and column rules, magic sums

Digits, Sudoku & Cryptarithm
4 questions
Digit logic and letter-to-number puzzles

Pyramids, Clock & Logic
5 questions
Pyramids, clocks, calendar and jug puzzles

Number Properties
5 questions
Multiples, remainders and digit products

Operators & Number Puzzles
3 questions
Custom operators, letter codes, sum and product

HOW TO USE

Give each question about 2 minutes, then check the answer and the solution below it.

Infosys Numerical Puzzle

Set 3 | 30 New Questions

1 Find the next term in the series: 2, 3, 6, 11, 18, 27, ?

A) 36
B) 38
C) 40
D) 42

Answer: B) 38

The differences are 1, 3, 5, 7, 9, so the next difference is 11. Hence 27 + 11 = 38.

2 Find the next term in the series: 3, 5, 6, 10, 9, 15, 12, ?

A) 18
B) 20
C) 19
D) 21

Answer: B) 20

Two patterns are interleaved: 3, 6, 9, 12 (+3 each time) and 5, 10, 15, 20 (+5 each time). The next term is 20.

3 Find the next term in the series: 1, 4, 27, 256, ?

A) 625
B) 1024
C) 3375
D) 3125

Answer: D) 3125

Each term is n raised to the power n: 1, $2^2 = 4$, $3^3 = 27$, 4 to the power 4 = 256. So the next is 5 to the power 5 = 3125.

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4 Find the next term in the series: 0, 6, 24, 60, 120, 210, ?

A) 320
B) 336
C) 330
D) 342

Answer: B) 336

Each term is $n^3 - n$: 0, 6, 24, 60, 120, 210. For n = 7: 343 - 7 = 336.

5 Find the next term in the series: 2, 4, 12, 48, 240, ?

A) 960
B) 1200
C) 1680
D) 1440

Answer: D) 1440

The multipliers are 2, 3, 4, 5 and then 6. So 240 × 6 = 1440.

6 Find the next term in the series: 1, 2, 4, 7, 11, 16, 22, ?

A) 29
B) 27
C) 28
D) 30

Answer: A) 29

The differences are 1, 2, 3, 4, 5, 6, so the next difference is 7. Hence 22 + 7 = 29.

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Set 3 | 30 New Questions

7 Find the next term in the series: 50, 49, 46, 41, 34, ?

A) 23
B) 25
C) 24
D) 26

Answer: B) 25

The differences are -1, -3, -5, -7, so the next difference is -9. Hence 34 - 9 = 25.

8 Find the next term in the series: 3, 7, 16, 35, 74, ?

A) 153
B) 147
C) 149
D) 155

Answer: A) 153

Each term is double the previous term plus an increasing number: +1, +2, +3, +4, +5. So 74 × 2 + 5 = 153.

9 Every row follows the same rule. Find the missing number.

3	4	13
5	2	27
6	7	?

A) 39
B) 43
C) 41
D) 47

Answer: B) 43

In each row, third = (first)² + second. So 6² + 7 = 36 + 7 = 43.

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10 Every column follows the same rule. Find the missing number.

1	2	3
2	3	4
3	7	?

A) 13
B) 11
C) 15
D) 17

Answer: A) 13

In each column, bottom = (top)² + middle. So 3² + 4 = 9 + 4 = 13.

11 This 3 × 3 grid is a magic square (all rows, columns and diagonals have the same sum). Find the number in the middle-right cell.

11	?	9
6	8	?
?	12	5

A) 8
B) 9
C) 11
D) 10

Answer: D) 10

The completed square is 11 4 9 / 6 8 10 / 7 12 5 with a magic sum of 24. Middle row: 24 - 6 - 8 = 10.

12 Every row follows the same rule. Find the missing number.

6	2	3
12	3	4
20	4	?

A) 4
B) 6
C) 5
D) 8

Answer: C) 5

In each row, first ÷ second = third. So 20 ÷ 4 = 5.

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13 Every row follows the same rule. Find the missing number.

5	3	12
7	4	24
9	5	?

A) 36
B) 45
C) 40
D) 48

Answer: C) 40

In each row, third = second × (first - 1). So 5 × (9 - 1) = 40.

14 Find the three-digit number that is exactly 18 times the sum of its digits.

A) 126
B) 144
C) 162
D) 198

Answer: C) 162

162 = 18 × 9 and its digit sum is 1 + 6 + 2 = 9. No other three-digit number satisfies this.

15 In the cryptarithm CROSS + ROADS = DANGER, each letter stands for a different digit. Find the value of D + A + N + G + E + R.

A) 29
B) 30
C) 33
D) 31

Answer: D) 31

96233 + 62513 = 158746. So D = 1, A = 5, N = 8, G = 7, E = 4 and R = 6, and their sum is 31.

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16 How many three-digit numbers have three different digits that add up to 6?

- A) 14
- B) 10
- C) 12
- D) 16

Answer: A) 14

The digit sets are {0, 1, 5} giving 4 numbers, {0, 2, 4} giving 4 numbers and {1, 2, 3} giving 6 numbers. Total = 14.

17 Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box contains the digits 1 to 4. Which digit replaces the ? in the last row?

1	?	?	4
?	4	1	?
3	?	4	?
?	2	?	1

- A) 1
- B) 2
- C) 3
- D) 4

Answer: D) 4

The completed rows are 1 3 2 4, 2 4 1 3, 3 1 4 2 and 4 2 3 1. The ? in the last row is 4.

18 In a number pyramid, each brick is the sum of the two bricks directly below it. The bottom row is 4, ?, 6 and the top brick is 40. Find the missing number.

- A) 10
- B) 12
- C) 15
- D) 18

Answer: C) 15

Let the missing number be x. The second row is 4 + x and x + 6, so the top is 2x + 10 = 40. Hence x = 15.

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19 What is the smaller angle between the hour hand and the minute hand of a clock at 7:20?

- A) 100
- B) 80
- C) 90
- D) 110

Answer: A) 100

At 20 minutes the minute hand is at 120°. The hour hand is at 210° + 10° = 220°. The angle is 220° - 120° = 100°.

20 If the 7th of a month is a Wednesday, which day of the week is the 30th of that month?

- A) Wednesday
- B) Thursday
- C) Friday
- D) Saturday

Answer: C) Friday

The 30th is 23 days after the 7th. 23 days = 3 weeks + 2 days, so Wednesday + 2 days = Friday.

21 You have a 5-litre jug and a 3-litre jug and unlimited water. Each fill, empty or pour counts as one step. What is the minimum number of steps needed to measure exactly 4 litres?

- A) 4
- B) 5
- C) 6
- D) 8

Answer: C) 6

Fill 5, pour into 3 (2 left), empty 3, pour the 2 into 3, fill 5, pour into 3 until full. That is 6 steps and leaves 4 litres in the 5-litre jug.

22 How many rectangles of all sizes (squares included) are there in a 3 × 3 square grid?

A) 27
B) 30
C) 42
D) 36

Answer: D) 36

A 3 × 3 grid has 4 horizontal and 4 vertical lines. Rectangles = C(4, 2) × C(4, 2) = 6 × 6 = 36.

23 What is the smallest four-digit number that is divisible by 5, 7 and 11?

A) 1100
B) 1210
C) 1265
D) 1155

Answer: D) 1155

The LCM of 5, 7 and 11 is 385. Its multiples are 385, 770 and 1155, so the smallest four-digit one is 1155.

24 What is the sum of all multiples of 7 between 1 and 100?

A) 700
B) 735
C) 728
D) 742

Answer: B) 735

The multiples are 7, 14, ... 98 (14 terms). Sum = 7 × (1 + 2 + ... + 14) = 7 × 105 = 735.

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25 What is the remainder when 3 raised to the power 100 is divided by 5?

- A) 2
- B) 3
- C) 1
- D) 4

Answer: C) 1

3 to the power 4 = 81 leaves remainder 1 when divided by 5. Since 100 = 4 × 25, the remainder is 1.

26 If x $ y = $x^2 + y^2$, what is the value of (2 $ 3) $ 1?

- A) 150
- B) 170
- C) 160
- D) 180

Answer: B) 170

2 $ 3 = $2^2 + 3^2 = 13$. Then 13 $ 1 = $13^2 + 1^2 = 169 + 1 = 170$.

27 If each letter is worth twice its alphabet position (A = 2, B = 4, ... Z = 52), what is the value of the word HEN?

- A) 54
- B) 50
- C) 52
- D) 56

Answer: A) 54

H = 16, E = 10 and N = 28. So the total is 16 + 10 + 28 = 54.

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28 Two numbers add up to 24 and their product is 143. What is the difference between the two numbers?

A) 2
B) 1
C) 3
D) 4

Answer: A) 2

The numbers are 11 and 13, since 11 + 13 = 24 and 11 × 13 = 143. The difference is 2.

29 How many two-digit numbers have a digit product of 12?

A) 4
B) 2
C) 3
D) 6

Answer: A) 4

The digit pairs with product 12 are (2, 6) and (3, 4). Each gives two numbers: 26, 62, 34 and 43. Total = 4.

30 The average of five consecutive odd numbers is 21. What is the largest of these numbers?

A) 23
B) 27
C) 25
D) 29

Answer: C) 25

The middle number is 21, so the numbers are 17, 19, 21, 23 and 25. The largest is 25.

Set 3 Answer Key

Q1	Q2	Q3	Q4	Q5
B	B	D	B	D
38	20	3125	336	1440

Q6	Q7	Q8	Q9	Q10
A	B	A	B	A
29	25	153	43	13

Q11	Q12	Q13	Q14	Q15
D	C	C	C	D
10	5	40	162	31

Q16	Q17	Q18	Q19	Q20
A	D	C	A	C
14	4	15	100	Friday

Q21	Q22	Q23	Q24	Q25
C	D	D	B	C
6	36	1155	735	1

Q26	Q27	Q28	Q29	Q30
B	A	A	A	C
170	54	2	4	25